

Descriptive statistics

DESCRIBING DATA BEFORE WE
INTERPRET IT

What are descriptive statistics?

- Descriptive statistics summarize data
- They help us answer:
 - How often did something happen?
 - What was typical?
 - How spread out were the scores?
- They do NOT test hypotheses by themselves



The three jobs of descriptive statistics

- Frequency: how often values appear
- Central tendency: what value is typical
- Spread: how much values differ

Start with raw data

Example: quiz scores

- 70, 72, 75, 75, 78, 80, 80, 80, 85, 88
- Raw data are useful, but hard to read quickly
- Descriptive stats make the pattern easier to see

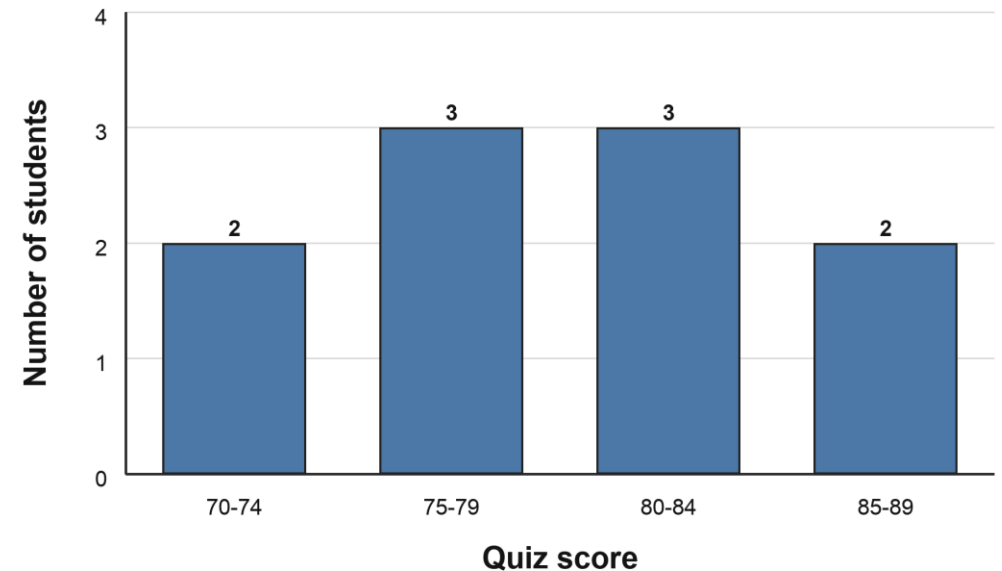
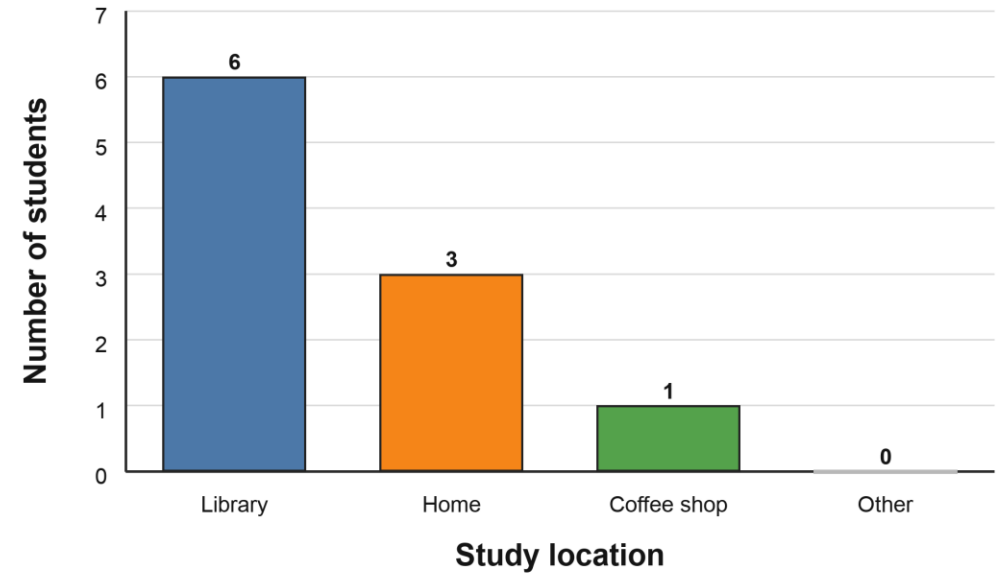
Frequency

- Frequency = how often each value ap
- Useful for:
 - quiz scores
 - survey responses
 - categories
 - Behaviors

Quiz score	Frequency
70	1
72	1
75	2
78	1
80	3
85	1
88	1

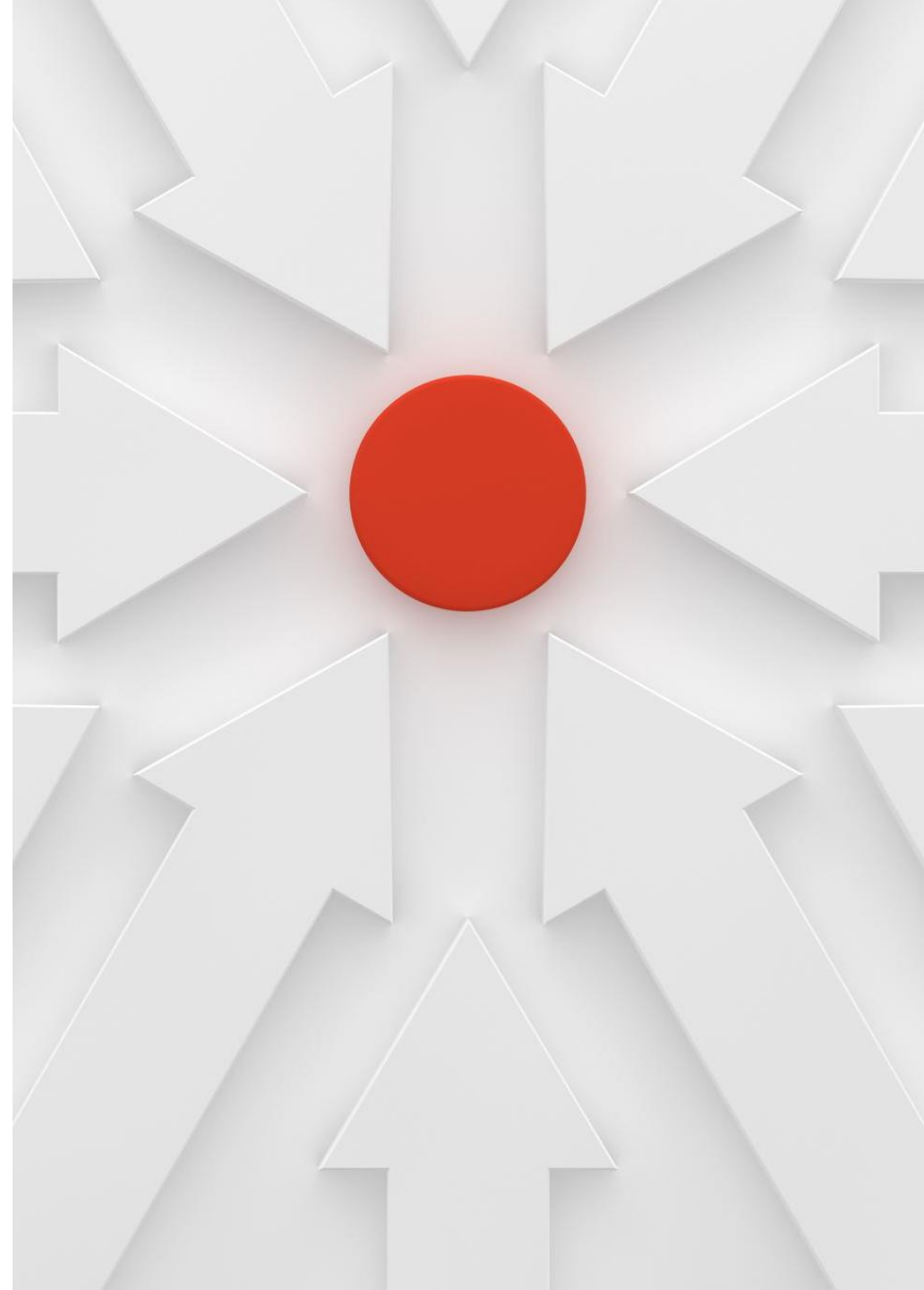
Graphing frequencies

- Frequency tables can become graphs
- Bar charts: good for categories
- Histograms: good for numerical values or ranges
- Always check the axes



Central tendency

- Central tendency = what value is typical?
- Three common measures:
 - mean
 - median
 - mode
- They answer similar questions in different ways



Central tendency: Mean

- Mean = arithmetic average
- Add all values, then divide by number of values
 - Quiz scores: $70 + 72 + 75 + 75 + 78 + 80 + 80 + 80 + 85 + 88 = 783$
 - 783 divided by $10 = 78.3$
 - Mean quiz score = 78.3
- Sensitive to extreme values



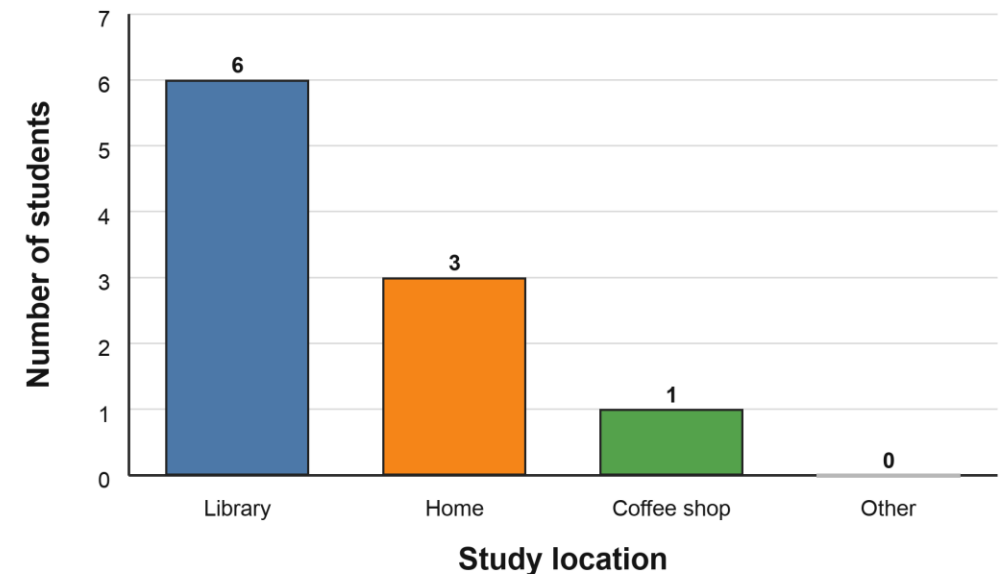
Central tendency: Median

- Median = middle value
- Put scores in order first
- Quiz scores: 70, 72, 75, 75, 78, 80, 80, 80, 85, 88
- Two middle scores: 78 and 80
- Median = 79

Central tendency: Mode

- Mode = most common value
- Quiz scores: 70, 72, 75, 75, 78, 80, 80, 80, 85, 88
- 80 appears three times
- Mode = 80
- Useful for categorical variables too

Quiz score	Frequency
70	1
72	1
75	2
78	1
80	3
85	1
88	1

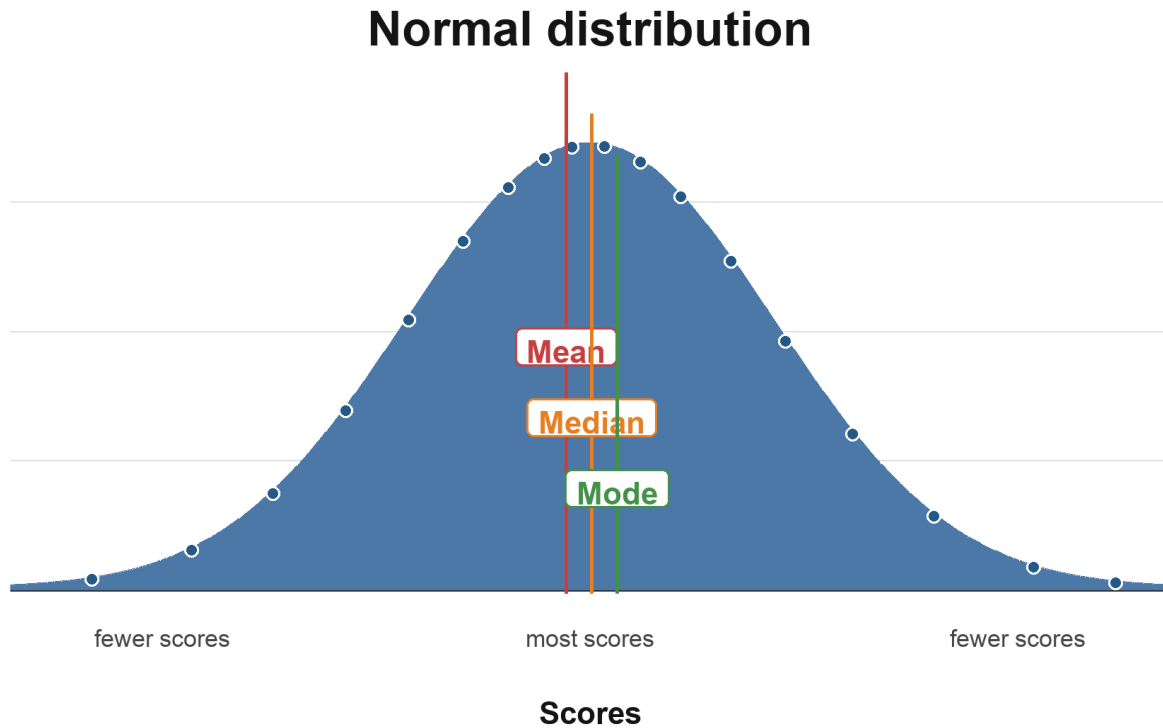


Distribution shape

A distribution
shows the
pattern of scores

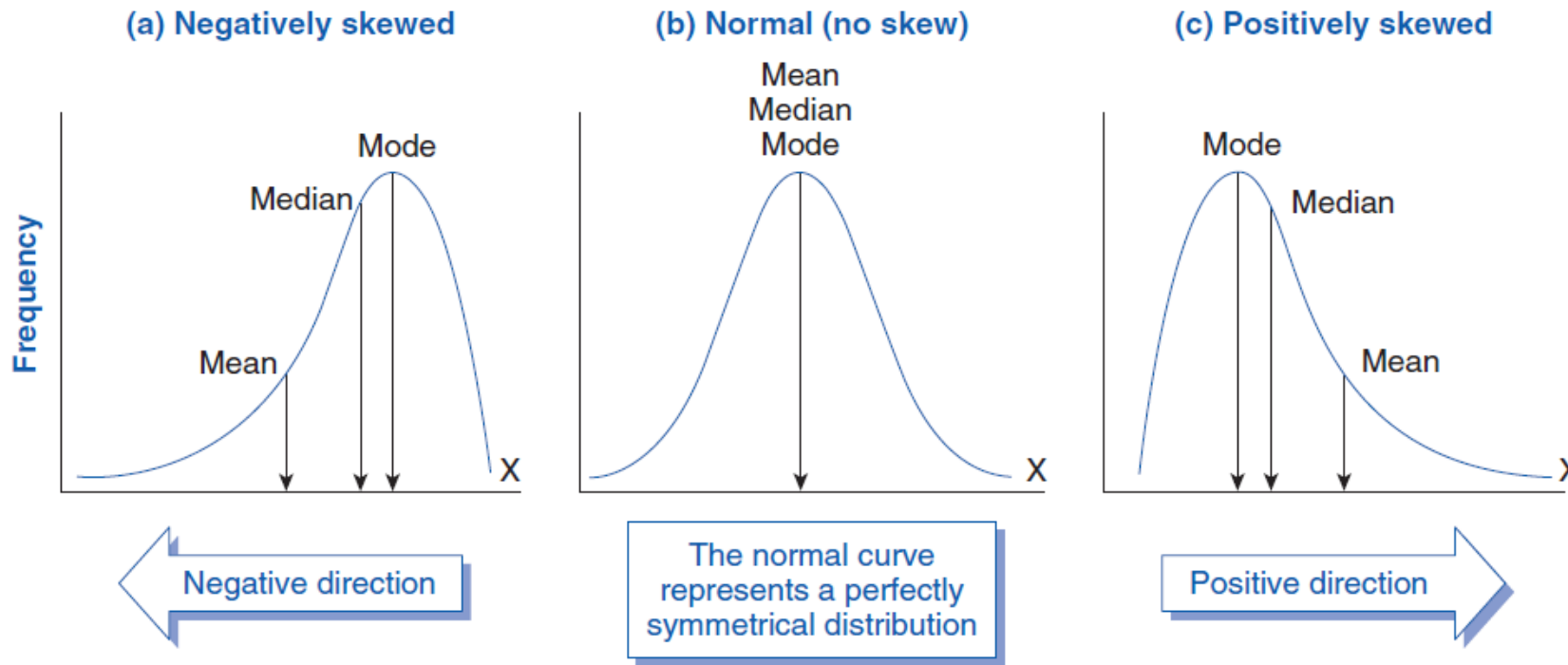
Shape affects
how we interpret
the average

Normal distribution



- Bell-shaped
- Most scores are near the center
- Fewer scores are very low or very high
- Mean, median, and mode are usually close together

Skewed distributions

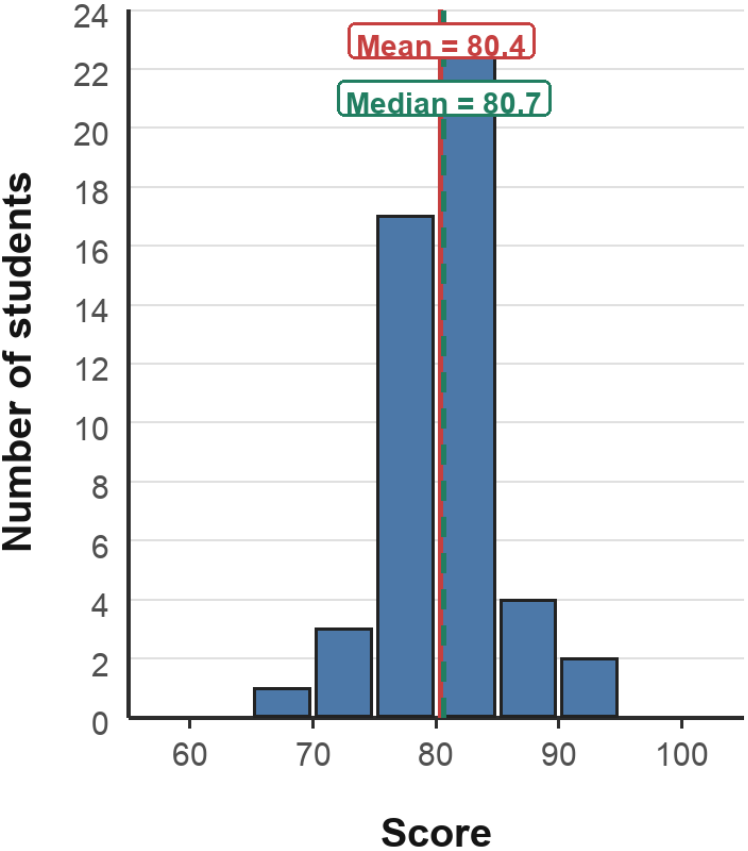


- Positive skew:
 - most scores are low
 - tail points toward higher values
- Negative skew:
 - most scores are high
 - tail points toward lower values
- Skew can pull the mean toward the tail

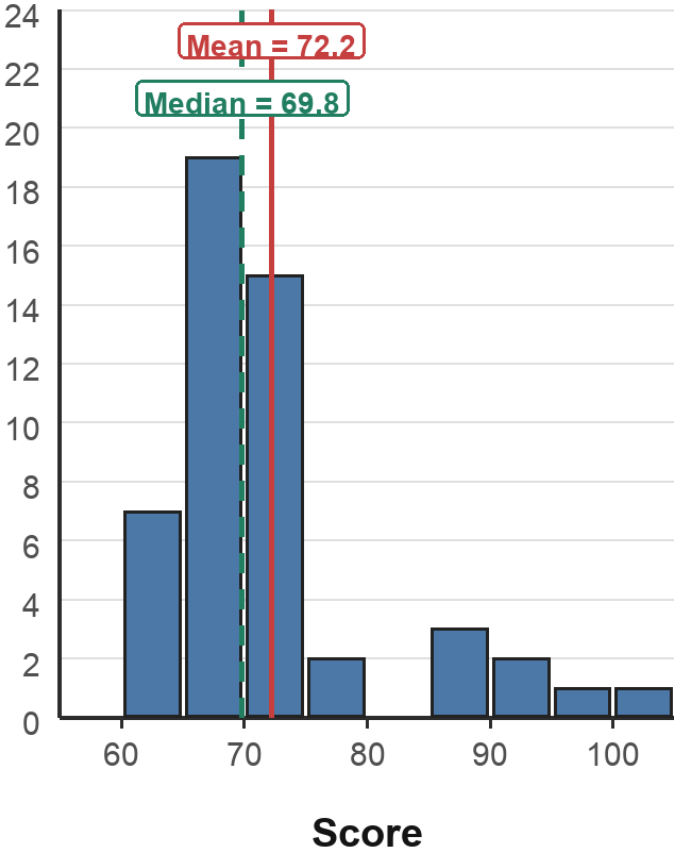
Comparing Distribution Shapes

— Mean - - - Median

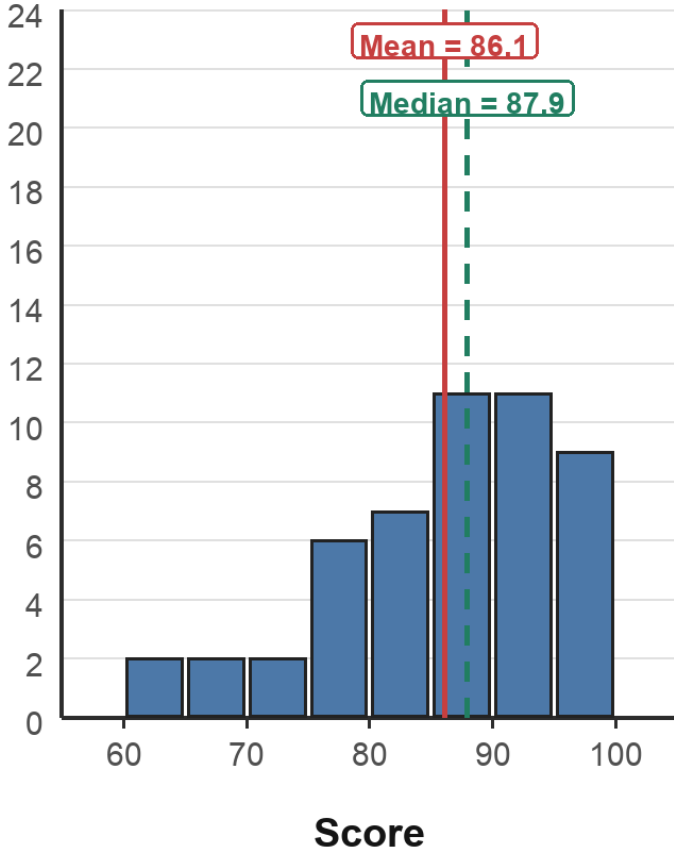
Normal Distribution



Positive Skew

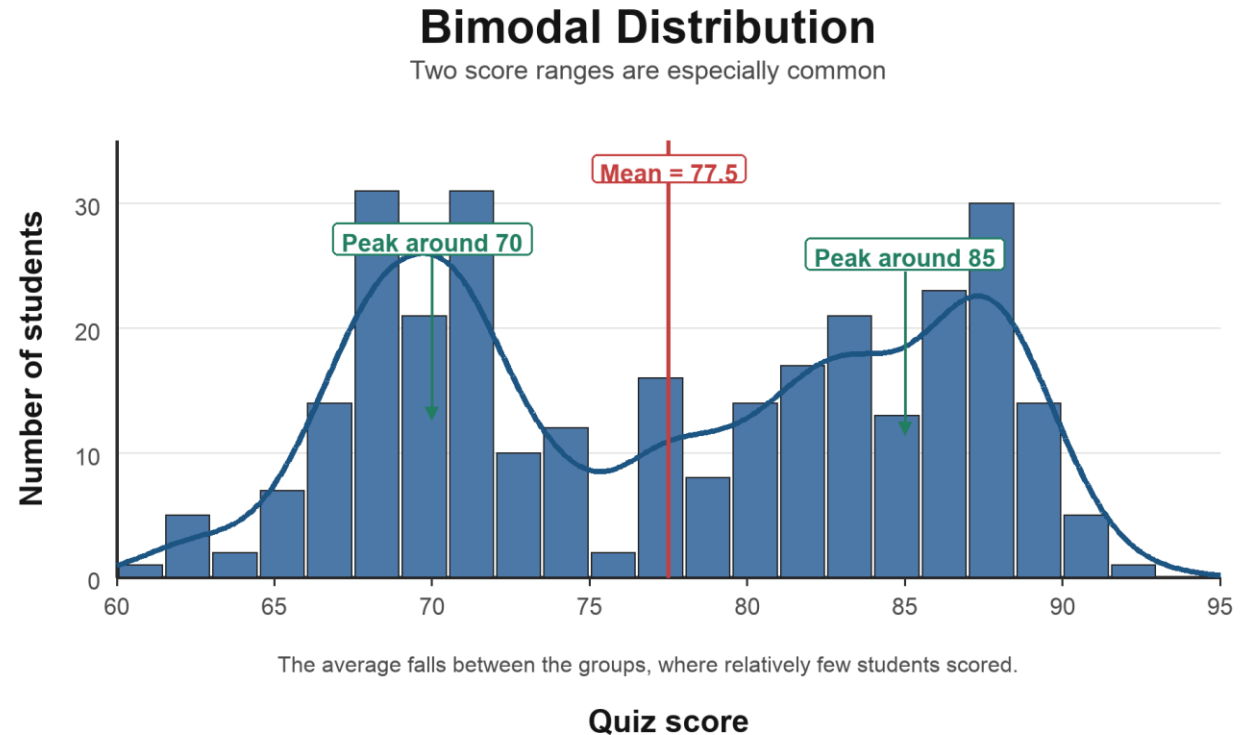


Negative Skew



Bimodal distributions

- Bimodal = two peaks
- Two values or ranges are especially common
- May suggest:
 - two different groups
 - an interesting split in the data

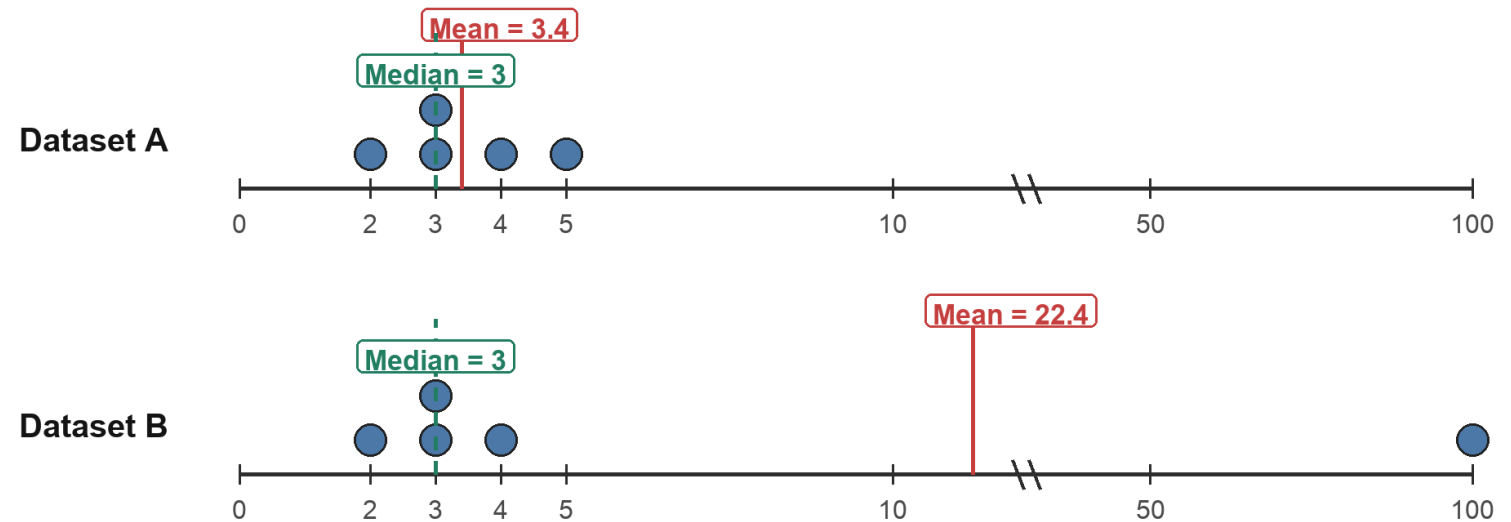


Same data, different summaries

Mean vs. Median With an Outlier

The median stays near the typical score; the mean gets pulled toward the outlier.

- Dataset A: 2, 3, 3, 4, 5
 - Mean = 3.4
 - Median = 3
 - Mode = 3
- Dataset B: 2, 3, 3, 4, 100
 - Mean = 22.4
 - Median = 3
 - Mode = 3



Same median, very different mean

— Mean - - - Median

Value



Attendance Activity

- For each variable, if you can only pick one descriptive statistic, what would you report?
 - Number of classes attended this semester
 - Whether a student attended today, yes or no
 - Total absences in a course
 - Attendance percentage with one extreme outlier



Spread

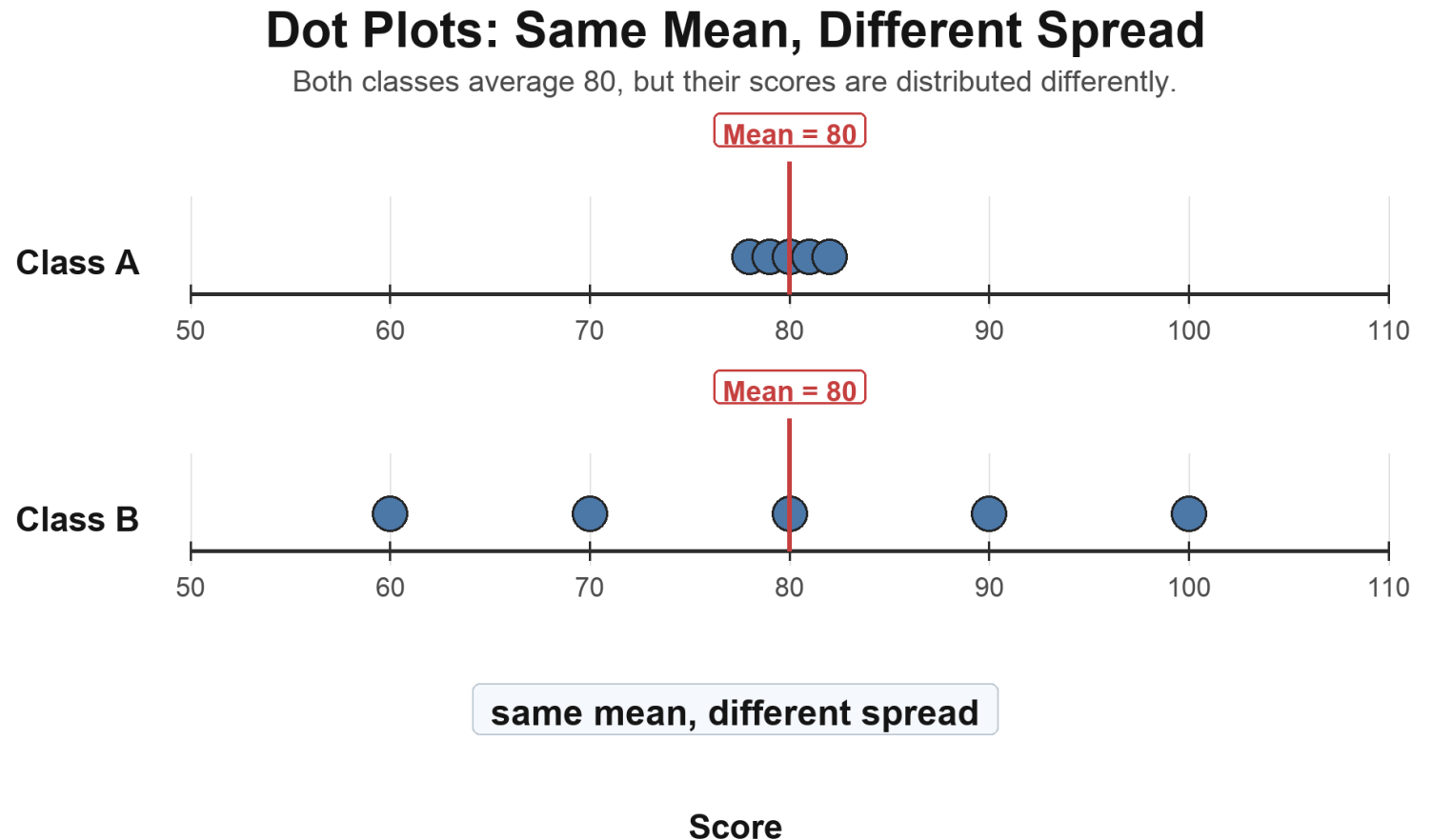
- Spread = how much scores differ
- Distribution = the overall pattern of scores
- Spread is one feature of a distribution
- Two datasets can have the same mean but different spread
- Common measures:
 - range
 - variance
 - standard deviation

Range

- Range = highest value minus lowest value
- Quiz scores:
 - lowest = 70
 - highest = 88
 - Range = 88 minus 70 = 18
- Easy to understand
- Very sensitive to outliers

Same mean, different spread

- Class A: 78, 79, 80, 81, 82
 - Mean = 80
 - Low variability
- Class B: 60, 70, 80, 90, 100
 - Mean = 80
 - High variability





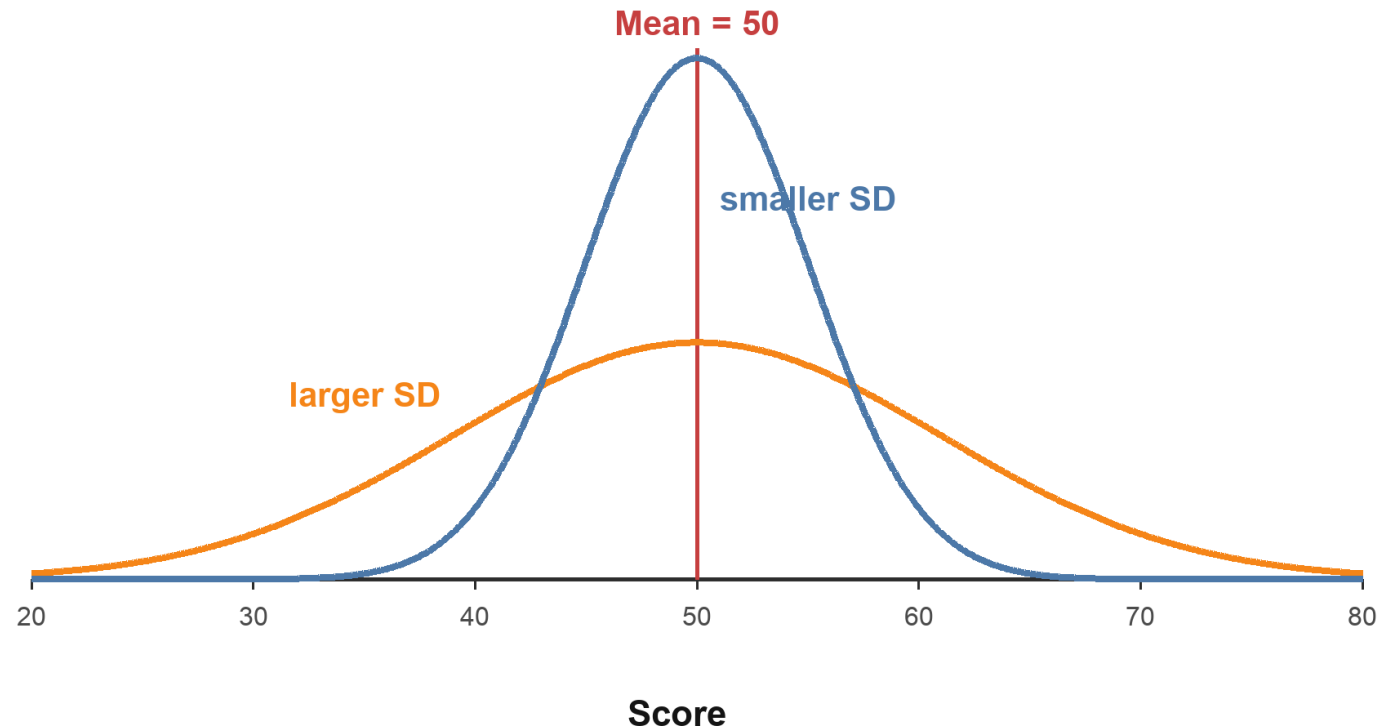
Variance

- Variance = average squared distance from the mean
- Conceptually: How far are scores from the average?
- Larger variance = more spread
- We focus more on the idea than the hand calculation

Standard deviation (SD)

Same Mean, Different Standard Deviation

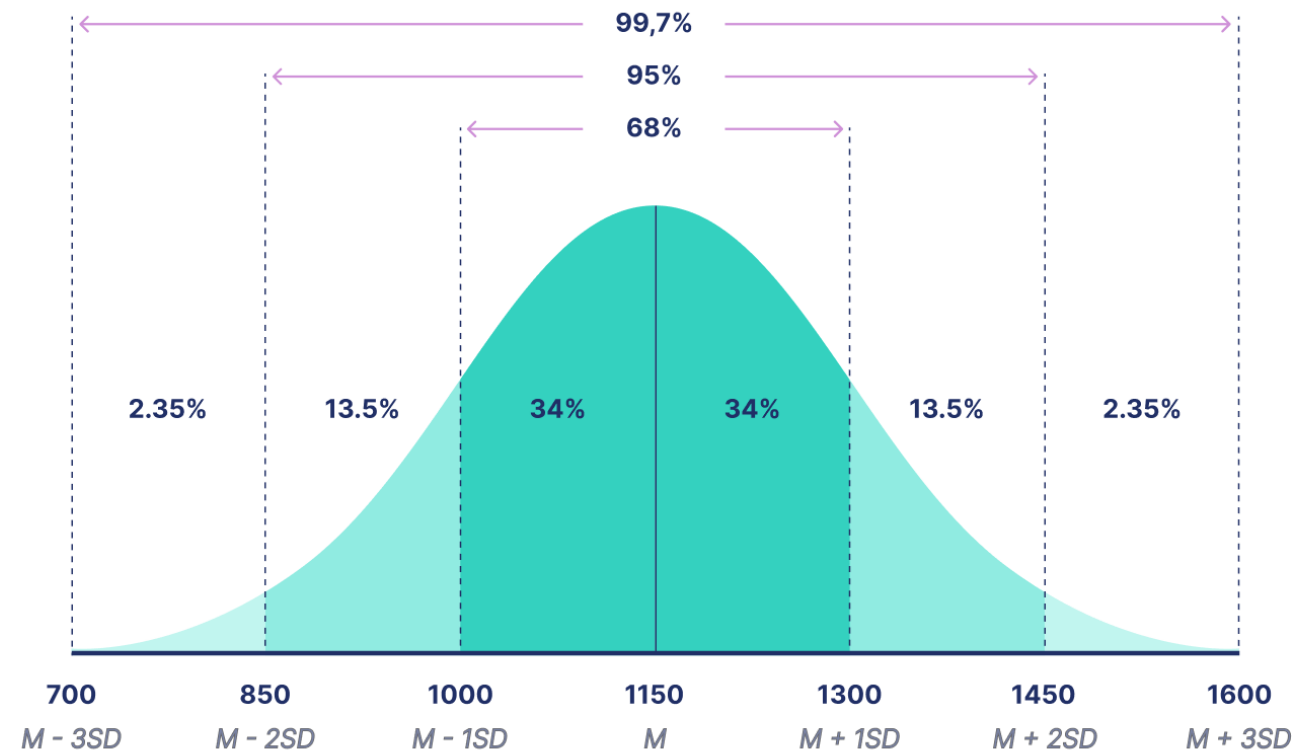
- Standard deviation = typical distance from the mean
- Based on variance
- Smaller SD: scores are close to the mean
- Larger SD: scores are more spread out
- Often reported with the mean



Standard deviation and normal distributions

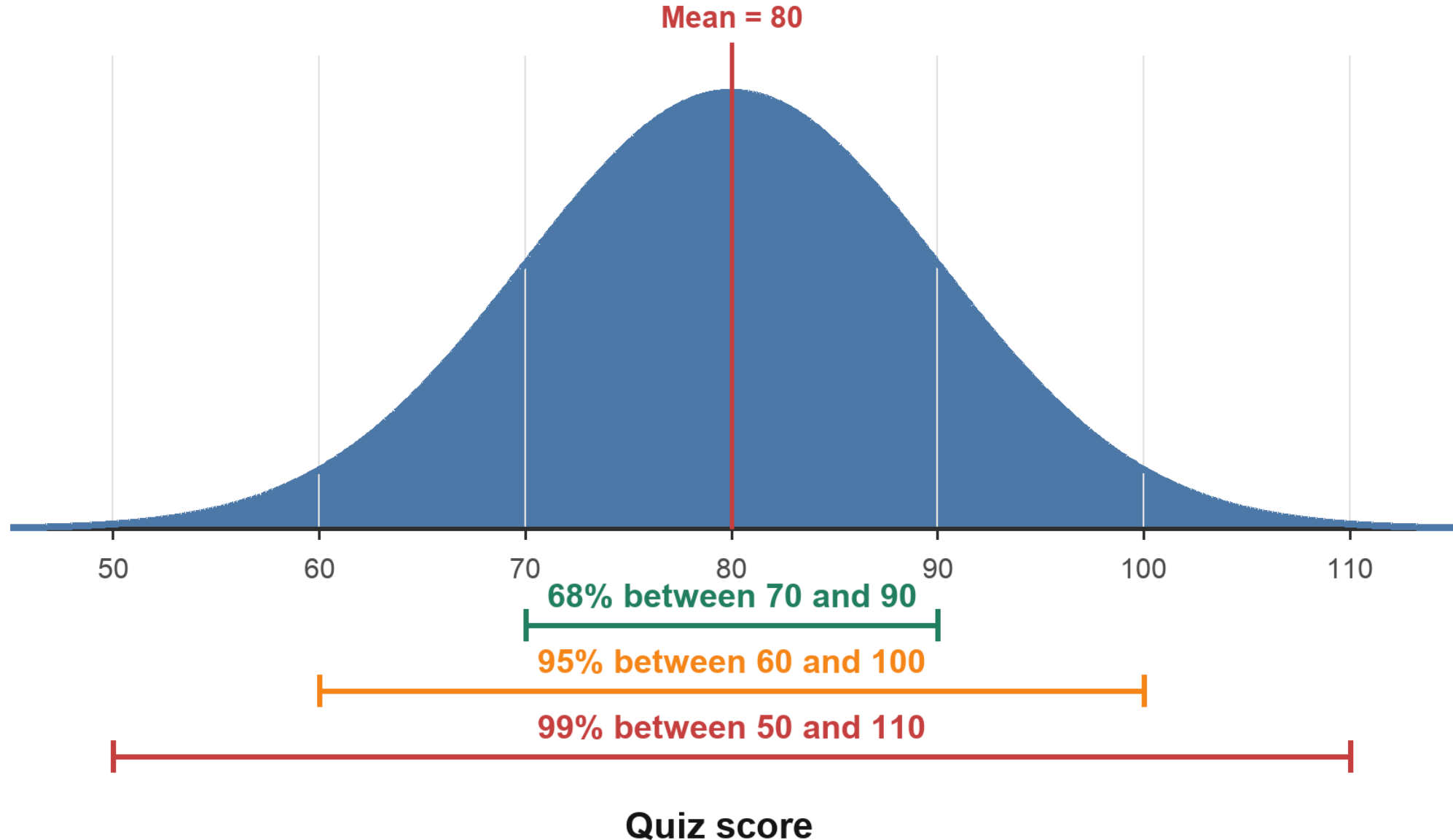
Using the empirical rule in a normal distribution

- In a normal distribution:
 - about 68 percent of scores fall within 1 SD
 - about 95 percent fall within 2 SDs
 - about 99 percent fall within 3 SDs
- Scores far from the mean are less common



Quiz Scores: Mean = 80, SD = 10

Most scores fall close to the mean; fewer scores fall farther away.



Outliers

- Outlier = an unusually low or high value
- Outliers can affect:
 - mean
 - range
 - variance
 - standard deviation
- Do not delete outliers automatically





Attendance Activity

A researcher asks 12 students: "How many hours did you sleep last night?"

Student responses: 4, 5, 5, 6, 6, 6, 7, 7, 8, 8, 9, 14

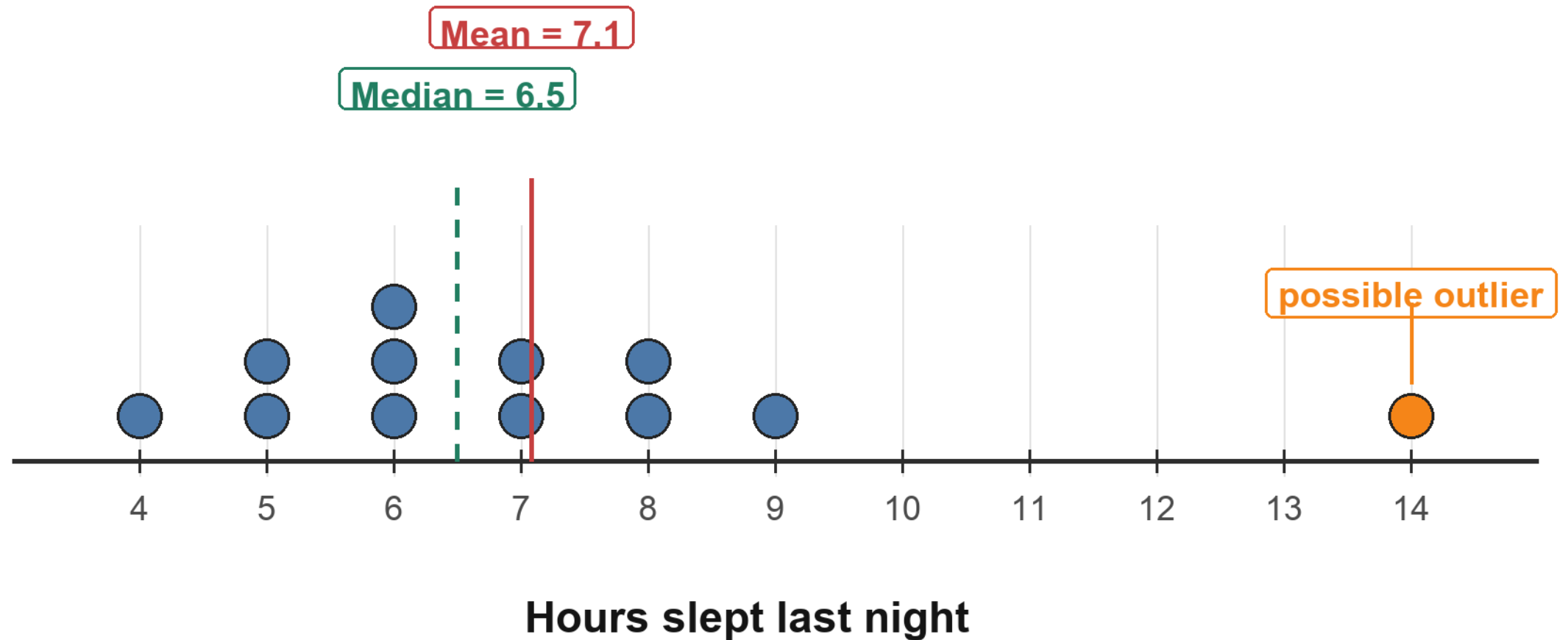
Answer the following questions

- Find the mean, median, mode, and range.
- What is the distribution shape? Roughly normal, positively skewed, negatively skewed, or bimodal?
- Is there a possible outlier?

Distribution of Sleep Hours

12 student responses to: "How many hours did you sleep last night?"

$n = 12$



Most students slept 5 to 8 hours, with one much higher response.